



SS2 PHYSICS LESSON NOTES (THIRD TERM)

WEEK 6: ENERGY QUANTIZATION

1. INTRODUCTION

In classical physics, energy was thought to vary continuously. However, early 20th-century discoveries in atomic and quantum physics revealed that energy, especially at the atomic and subatomic levels, exists in discrete packets, not continuous forms. This led to the concept of **energy quantization**, a fundamental idea in quantum physics that radically transformed our understanding of atomic structure and electromagnetic radiation.

2. DEFINITION OF ENERGY QUANTIZATION

Energy quantization refers to the concept that energy levels of electrons in an atom are **not continuous**, but rather exist in **discrete (quantized) energy states**.

This means that:

An electron in an atom can only possess certain specific energy values and not just any arbitrary amount.

These energy states are associated with orbitals or energy levels (shells) around the nucleus.

An electron must absorb or emit a **fixed amount of energy**, called a **quantum**, to move from one energy level to another.

3. DISCRETE ENERGY LEVELS IN ATOMS



Atoms have a set of fixed energy levels, denoted by **quantum numbers ($n = 1, 2, 3, \dots$)**. These energy levels are like rungs of a ladder — an electron can "stand" on one rung or another, but cannot exist between rungs.

Term	Definition
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Ground state	The lowest energy level ($n = 1$), where the electron is most stable
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Excited state	Any energy level higher than the ground state ($n = 2, 3, \dots$), where the electron has gained energy
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Each level corresponds to a specific amount of energy, and the difference between any two levels represents the energy an electron must absorb or emit during a transition.

4. ELECTRON TRANSITIONS AND QUANTA OF ENERGY

Electrons can move between energy levels by either:

Process	Description
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Absorbing energy	Moving to a higher energy level (excitation)
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Emitting energy	Falling to a lower energy level (de-excitation)
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The energy exchanged during these transitions is in the form of **electromagnetic radiation** (like light), and it is always a **discrete quantum of energy**, not a continuous flow.

Formula:

$$\Delta E = E_{\text{higher}} - E_{\text{lower}} = hf$$

Where:

ΔE = energy absorbed/emitted (in joules)

h = Planck's constant (6.626×10^{-34} Js)

f = frequency of the absorbed/emitted radiation (in Hz)

5. PLANCK'S HYPOTHESIS AND DERIVATION OF $E = hf$

In **1900**, **Max Planck** introduced the idea that electromagnetic energy is quantized.

Planck's Hypothesis:

"Energy is emitted or absorbed in discrete units or quanta, and the energy E of each quantum is directly proportional to the frequency f of radiation."

Mathematical Expression:



$$E = hf$$

Where:

E = energy of a single quantum of radiation (in joules)

h = Planck's constant (6.626×10^{-34} Js)

f = frequency of radiation (in hertz, Hz)

This formula marked the beginning of **quantum theory**, breaking away from classical physics, which treated energy as continuous

6. THE PHOTOELECTRIC EFFECT

The **photoelectric effect** is a phenomenon where electrons are ejected from the surface of a metal when light of **sufficient frequency** shines on it.

Experimental Observations:

Observation

Description

1

No electrons are emitted unless the light's frequency is above a **threshold frequency (f_0)**, regardless of intensity

2

The **kinetic energy** of emitted electrons depends on the **frequency** of the light, not its intensity



Observation	Description
3	The number of electrons emitted depends on the intensity (brightness) of the light, if frequency is above threshold
4	The emission of electrons is instantaneous when the threshold frequency is reached

Einstein's Explanation:

These observations could not be explained by classical wave theory but were successfully explained by **Einstein** using Planck's quantum hypothesis.

Einstein proposed that light is made up of particles called **photons**, each carrying energy **E = hf**. When a photon strikes an electron, it transfers energy.

Einstein's Photoelectric Equation:

$$hf = \phi + \frac{1}{2}mv^2$$

Where

ϕ (phi) = work function (minimum energy needed to free an electron)

$\frac{1}{2}mv^2$ = kinetic energy of the emitted electron



This confirmed that energy is quantized, further validating Planck's idea

7. QUANTIZATION AND SPECTRAL LINES

When electrons in an atom move between energy levels, they emit or absorb specific amounts of energy. These energy changes correspond to particular frequencies or wavelengths of light.

Each element emits light at **specific frequencies**, producing a **line spectrum** (as opposed to a continuous spectrum). This is known as an **atomic emission** or **absorption spectrum**.

Example:

The **hydrogen atom** emits visible light at specific lines in the **Balmer series**, corresponding to electron transitions to the **$n = 2$** level.

These spectral lines serve as "**fingerprints**" of **elements**, used in chemical analysis and astronomy.

8. SIGNIFICANCE OF ENERGY QUANTIZATION

Significance

Explanation

Explains stability of atoms

Electrons cannot spiral into the nucleus as they can only occupy fixed levels



Significance	Explanation
Accounts for discrete spectral lines	Explains emission and absorption spectra
Supports key phenomena	Explains photoelectric effect, blackbody radiation, and Bohr's atomic model
Forms basis for modern technology	Foundation for quantum mechanics, modern electronics, lasers, and semiconductors

9. SUMMARY OF KEY FORMULAS

Formula	Name	Variables
$\Delta E = hf$	Energy of a quantum	ΔE = energy change, h = Planck's constant, f = frequency
$hf = \phi + \frac{1}{2}mv^2$	Einstein's photoelectric equation	ϕ = work function, $\frac{1}{2}mv^2$ = kinetic energy

Constants:



$h = 6.626 \times 10^{-34} \text{ Js}$ (Planck's constant)

$c = 3.0 \times 10^8 \text{ m/s}$ (speed of light)

10. EVALUATION (WAEC/NECO PAST QUESTIONS)

Question 1 (WAEC 2017, Q7)

The energy of a photon is $3.0 \times 10^{-19} \text{ J}$. Calculate the frequency of the radiation. ($h = 6.63 \times 10^{-34} \text{ Js}$)

Question 2 (NECO 2015, Q5)

Explain why light of frequency below a certain value does not cause photoemission from a metal surface.

Question 3 (WAEC 2016, Q4)

Describe the photoelectric effect and explain how it supports the quantum theory of light.

Question 4 (WAEC 2014, Q3b)

Define the term energy quantization and explain its role in the emission of line spectra.



Question 5 (NECO 2020, Q6)

Derive the relationship $E = hf$ and state its significance.

11. ASSIGNMENT

Define the following terms:

- (a) Energy quantization
- (b) Photon
- (c) Threshold frequency
- (d) Work function

An electron in an atom absorbs 4.14×10^{-19} J of energy. Calculate the frequency of the absorbed radiation. ($h = 6.63 \times 10^{-34}$ Js)

Discuss how the photoelectric effect cannot be explained by classical physics but supports quantum theory.

Explain why the hydrogen atom emits light of specific wavelengths.

Distinguish between continuous spectrum and line spectrum with examples.

